AI-POWERED PRODUCT MANAGEMENT AND GROWTH MARKETING

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CAPSTONE PROJECT REPORT

ENHANCING CHARGING ACCESSIBILITY AND CUSTOMER EXPERIENCE THROUGH PRODUCT INNOVATION FOR ATHER ENERGY

(*Proposed Product Solutions for Strengthening the Ather Grid Charging Ecosystem)*

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ABSTRACT

The Indian electric two-wheeler market is witnessing rapid growth as consumers increasingly adopt electric mobility due to rising fuel prices, government incentives, and growing environmental awareness. However, despite significant improvements in battery technology and vehicle performance, one of the primary barriers to widespread EV adoption continues to be charging accessibility and range anxiety. Customers often hesitate to undertake longer journeys because of uncertainty regarding charger availability, charging time, and the reliability of public charging infrastructure. These concerns directly influence purchase decisions, customer satisfaction, and long-term brand loyalty.

Ather Energy has emerged as one of India's leading premium electric vehicle manufacturers by creating an integrated ecosystem consisting of smart electric scooters, intelligent software, connected mobile applications, and the Ather Grid fast-charging network. Unlike many competitors, Ather has invested heavily in building its own charging infrastructure while simultaneously introducing home charging solutions and interoperable charging standards such as LECCS (Light Electric Combined Charging System) to promote a unified EV charging ecosystem. Through strategic partnerships with OEMs and Charge Point Operators (CPOs), Ather has positioned itself as a technology leader committed to accelerating India's transition toward sustainable transportation.

Although Ather currently offers multiple charging solutions including Home Chargers, neighborhood Chargers, and Ather Grid Fast Chargers, there remain several opportunities to improve the overall charging experience. Urban apartment residents often lack dedicated home charging facilities, public charging stations may not always be conveniently located along frequently travelled routes, and users requiring emergency charging can face uncertainty regarding charger availability during peak hours. These challenges contribute to range anxiety, inefficient trip planning, and reduced confidence in adopting electric mobility for longer or more frequent journeys.

This project adopts a Product Management (PM) approach to analyze Ather Energy's existing charging ecosystem from both business and customer perspectives. The study

begins by evaluating the company's current charging infrastructure, customer charging behaviour, and market positioning before identifying key pain points affecting user experience. The project further benchmarks Ather's charging ecosystem against industry practices and explores opportunities to strengthen its competitive advantage through technology-driven and customer-centric innovations.

The proposed solutions focus on enhancing both infrastructure expansion and digital product capabilities. A key recommendation is the creation of a dedicated cross-functional Charging Intelligence Team comprising data analysts, software engineers, product managers, and operations specialists. This team would leverage anonymized charging data from the Ather application including home locations, charging frequency, travel routes, and public charging usage to identify underserved regions where additional Neighborhood Chargers should be deployed. Such a data-driven infrastructure planning approach ensures optimal investment while addressing the challenges faced by residents of apartments, gated communities, and workplaces without private charging access.

To further reduce range anxiety, the project proposes an intelligent "Plan Charging on the Way" feature within the Ather mobile application. By integrating battery state-of-charge, real-time navigation, estimated energy consumption, terrain conditions, and charger availability, the application can automatically recommend optimal charging stops before the rider begins a journey. Rather than requiring users to manually search for charging stations, the system proactively plans to charge breaks, thereby increasing travel confidence and improving overall user convenience during intercity and long-distance rides.

Another major enhancement involves introducing a Premium Fast-Charging Slot Reservation System for high-demand Ather Grid locations. Through this feature, users can reserve charging slots of predefined durations by paying a nominal premium fee, ensuring charger availability during urgent travel situations. Intelligent slot management, automated cancellation windows, and dynamic utilization of reserved chargers during off-peak periods would maximize infrastructure efficiency while significantly reducing waiting times and improving customer satisfaction.

Recognizing that charging infrastructure remains the backbone of electric mobility adoption, the project also recommends accelerating Ather Grid expansion through strategic partnerships with fuel retail companies such as Indian Oil Corporation (IOC), Bharat Petroleum (BPCL), and Hindustan Petroleum (HPCL). Leveraging the nationwide presence of petrol stations across Tier I, Tier II, and Tier III cities would enable rapid deployment of reliable charging infrastructure while supporting Ather's long-term vision of becoming India's largest interoperable two-wheeler charging network. Additionally, continued collaboration with OEMs adopting the LECCS charging standard would further strengthen network interoperability and improve charging accessibility for a broader EV user base.

To improve customer retention and encourage greater utilization of the charging ecosystem, this project proposes the introduction of an Ather Charging+ Subscription. The subscription would offer benefits such as unlimited or discounted charging within selected plans, priority charging access, exclusive reservation privileges, and value-added services tailored for daily commuters, delivery partners, fleet operators, and gig economy workers. Such a recurring revenue model has the potential to strengthen customer loyalty while improving utilization of the expanding charging infrastructure.

Finally, the project recommends a data-driven marketing strategy that utilizes geospatial demand analysis to identify regions with high EV adoption potential but limited charging infrastructure. Targeted digital campaigns, local business collaborations, and investor outreach initiatives can attract commercial property owners, residential societies, and entrepreneurs interested in hosting new charging stations, thereby accelerating ecosystem growth with lower capital expenditure.

Overall, this project aims to demonstrate how product management principles, data analytics, intelligent software features, strategic partnerships, and customer-centric service innovations can collectively transform Ather's charging ecosystem. By reducing charging uncertainty, improving infrastructure accessibility, enhancing digital planning capabilities, and introducing new business models, the proposed solutions seek to minimize range anxiety, increase customer confidence, strengthen Ather's competitive positioning, and contribute toward the broader adoption of electric mobility across

India. The recommendations presented in this report are designed not only to improve the charging experience for existing Ather customers but also to establish a scalable and future-ready charging ecosystem capable of supporting the next generation of connected electric transportation.

1. COMPANY PROFILE- ATHER ENERGY

1.1 Introduction

Ather Energy is one of India's leading electric vehicles (EV) manufacturers, recognized for pioneering smart, connected, and performance-oriented electric scooters. Established in 2013, the company has played a significant role in transforming India's two-wheeler EV industry by integrating advanced battery technology, intelligent software, cloud connectivity, and a rapidly expanding charging infrastructure into a single ecosystem.

Unlike traditional automobile manufacturers that transitioned from internal combustion engine (ICE) vehicles to electric mobility, Ather was founded as an EV-first company with a strong focus on innovation, research, and technology. The company develops most of its critical components in-house, including battery packs, Battery Management Systems (BMS), vehicle control software, dashboards, mobile applications, and charging infrastructure. This vertically integrated approach enables Ather to deliver a seamless user experience while maintaining high standards of quality, safety, and performance.

Over the years, Ather has expanded from a startup into one of India's fastest-growing premium EV brands. Its product portfolio currently includes the Ather 450X, Ather 450S, and the family-oriented Ather Rizta, catering to different customer segments while maintaining the company's core philosophy of intelligent mobility.

Beyond manufacturing electric scooters, Ather has built one of India's largest public charging networks for electric two-wheelers, known as Ather Grid, which has become a key differentiator for the brand. The company continues to invest heavily in charging infrastructure, connected services, software innovation, and customer experience,

positioning itself as a complete electric mobility ecosystem rather than simply a vehicle manufacturer.

1.2 Company Overview

Particular	**Details**
Company Name	Ather Energy Limited
Founded	2013
Founders	Tarun Mehta & Swapnil Jain
Headquarters	Bengaluru, Karnataka, India
Industry	Electric Vehicle Manufacturing
Business Type	Electric Two-Wheeler Manufacturer
Products	Electric Scooters, Charging Infrastructure, Connected Mobility Solutions
Flagship Products	Ather 450X, Ather 450S, Ather Rizta
Charging Network	Ather Grid
Mobile Platform	Ather App
Technology	Smart Connected Electric Mobility

1.3 Company History

Ather Energy was founded by Tarun Mehta and Swapnil Jain, graduates of the Indian Institute of Technology (IIT) Madras. While studying electric mobility and sustainable transportation, they identified several shortcomings in the existing electric scooter market, including poor performance, outdated technology, unreliable batteries, and limited charging infrastructure.

Rather than modifying existing scooters, they envisioned designing a completely new electric vehicle from the ground up—one that would offer premium performance, intelligent software, advanced connectivity, and a superior ownership experience.

In its early years, Ather focused heavily on research and development, investing significant resources in battery engineering, electronics, software systems, and manufacturing capabilities. The company officially launched the Ather 450 in 2018, introducing features that were uncommon in the Indian two-wheeler market, such as a touchscreen dashboard, cloud connectivity, navigation, Over-the-Air (OTA) software updates, and fast charging support.

Since then, Ather has continuously expanded its product lineup, manufacturing facilities, charging network, and retail presence across India. Today, the company serves customers in numerous cities while continuing to invest in technology, sustainability, and infrastructure development.

1.4 Vision

"To build India's most intelligent and sustainable mobility ecosystem by accelerating the transition toward electric transportation."

Ather envisions a future where electric mobility becomes the preferred mode of transportation through technological innovation, reliable charging infrastructure, and customer-centric solutions.

1.5 Mission

Ather Energy aims to:

- Accelerate the adoption of electric mobility across India.
- Deliver premium, intelligent, and connected electric vehicles.
- Build a reliable and accessible nationwide charging infrastructure.
- Continuously innovate through software-driven vehicle development.
- Reduce dependence on fossil fuels and contribute toward environmental sustainability.
- Enhance customer experience through seamless integration of vehicles, charging, and digital services.

1.6 Core Values

Ather's growth is guided by several core principles that define its products, culture, and long-term strategy.

Innovation

The company continuously invests in research and development to improve battery technology, vehicle software, charging systems, and connected services.

Customer-Centricity

Every product feature is designed to improve convenience, safety, performance, and the overall ownership experience.

Sustainability

Ather promotes environmentally responsible transportation by reducing carbon emissions and supporting India's clean energy transition.

Quality and Reliability

Strict quality control, rigorous testing, and in-house engineering ensure high product reliability and long-term durability.

Continuous Improvement

Through OTA software updates and data-driven product enhancements, Ather continuously improves its vehicles even after purchase.

2. PRODUCT PORTFOLIO

2.1 Business Model

Ather follows an integrated business model that combines manufacturing, software development, charging infrastructure, retail operations, and after-sales service.

The company's revenue is generated through multiple channels:

- Electric scooter sales
- Charging infrastructure services
- Accessories and merchandise
- Extended warranty packages
- Connected software services
- Subscription-based offerings
- Service and maintenance

Unlike conventional manufacturers that rely primarily on dealership networks, Ather operates **Experience Centers**, where customers can explore products, take test rides, purchase vehicles, and receive after-sales support under one ecosystem.

Ather has developed electric scooters targeted at different customer segments.

Ather 450X

The flagship performance scooter is designed for technology enthusiasts and urban commuters. It offers high acceleration, multiple ride modes, connected navigation, fast charging compatibility, and premium features.

Ather 450S

A comparatively affordable version of the 450 series, designed to make premium electric mobility accessible while retaining smart connectivity and everyday usability.

Ather Rizta

A family-oriented electric scooter launched to cater to customers seeking comfort, practicality, larger seating capacity, and increased storage while maintaining Ather's connected technology ecosystem.

2.2 Technology Ecosystem

One of Ather's biggest competitive advantages is its focus on developing technology in-house rather than depending heavily on third-party suppliers.

Ather App

The Ather mobile application enables riders to:

- Monitor battery percentage
- Track charging progress
- Locate nearby charging stations
- View ride history
- Access trip statistics
- Schedule service appointments
- Receive software update notifications
- Monitor vehicle health

Connected Dashboard

Every premium Ather scooter features an intelligent dashboard that provides:

- Turn-by-turn navigation
- Live traffic information

- Call and notification alerts
- Ride modes
- Battery status
- Range estimation
- Charging information

This integrated approach allows Ather to rapidly introduce new features, improve battery performance, diagnose issues remotely, and provide software updates throughout the vehicle's lifecycle.

2.3 Charging Infrastructure- Ather Grid

Recognizing that charging infrastructure is essential for EV adoption, Ather established **Ather Grid**, one of India's largest publics fast-charging networks for electric two-wheelers.

The network provides:

- DC Fast Charging
- Route-based charging support
- Urban charging accessibility
- Highway charging expansion
- Smart charger discovery through the Ather App
- Real-time charger availability
- Interoperable charging using LECCS standards

Ather Grid significantly reduces range anxiety by enabling users to quickly recharge their scooters during longer journeys.

The company continues to expand this network through strategic collaborations with commercial establishments, residential communities, workplaces, malls, hotels, cafés, and fuel station operators.

2.4 Competitive Advantages

Ather differentiates itself from competitors through several strategic strengths:

Integrated Ecosystem

Ather offers an end-to-end electric mobility ecosystem consisting of vehicles, charging infrastructure, connected applications, and after-sales support.

Software-Defined Vehicles

Continuous Over-the-Air (OTA) updates ensure vehicles receive new features, performance improvements, bug fixes, and security enhancements without requiring dealership visits.

Premium User Experience

Touchscreen dashboards, Google Maps navigation, ride analytics, remote diagnostics, and intelligent battery management provide superior ownership experience.

Expanding Charging Network

The Ather Grid network addresses one of the biggest barriers to EV adoption by improving public charging accessibility across major cities and highways.

Data-Driven Decision Making

Connected vehicles generate valuable insights into charging behaviour, battery performance, ride patterns, and customer usage, enabling continuous product improvement.

2.5 Market Position

India's electric two-wheeler market has become increasingly competitive with the entry of manufacturers such as Ola Electric, TVS Motor, Bajaj Auto, Hero MotoCorp (Vida), and others. Despite this competition, Ather has established itself as a premium technology-focused brand known for quality, innovation, and reliability rather than competing solely on price.

Its emphasis on connected mobility, intelligent software, high-performance scooters, and a comprehensive charging ecosystem has enabled the company to build strong brand loyalty among urban commuters and technology-conscious consumers.

2.6 Relevance to This Project

For an electric vehicle company, charging infrastructure is not merely a supporting service it is a critical component of the overall customer experience. The availability, accessibility, reliability, and convenience of charging directly influence customer satisfaction, purchase decisions, and long-term adoption of electric mobility.

As Ather continues to expand its customer base across metropolitan, suburban, and emerging markets, further strengthening its charging ecosystem becomes increasingly important. Enhancing infrastructure planning through data analytics, introducing intelligent route-based charging recommendations, enabling charging slot reservations, expanding strategic partnerships, and launching subscription-based charging services can significantly reduce range anxiety while improving operational efficiency and customer retention.

This project focuses on identifying these opportunities and proposing scalable product management solutions that align with Ather Energy's vision of building India's most intelligent, accessible, and future-ready electric mobility ecosystem.

3. PROBLEM STATEMENT

3.1 Background

India's electric two-wheeler market is experiencing rapid growth, driven by rising fuel prices, increasing environmental awareness, supportive government policies, and advancements in battery technology. However, despite these positive trends, charging infrastructure continues to be one of the biggest barriers to mass EV adoption.

While Ather Energy has invested significantly in building the Ather Grid network and offers multiple charging solutions including Home Chargers, neighborhood Chargers,

and Fast Chargers many customers still experience uncertainty regarding charging availability, especially during long-distance travel and in densely populated urban areas. This uncertainty, commonly referred to as range anxiety, affects customer confidence, purchase decisions, and overall satisfaction.

Unlike petrol scooters that can be refueled within minutes at thousands of fuel stations, electric scooters require users to plan charging based on battery state, charger availability, and travel distance. As EV adoption increases, customer expectations are shifting from simply having charging stations available to expecting a predictable, intelligent, and seamless charging experience.

3.2 Existing Challenges

1. Uneven Distribution And limited availability Of Charging Infrastructure

Although Ather Grid is expanding rapidly, charging stations remain concentrated in metropolitan cities and premium commercial locations. Several residential neighborhoods, suburban areas, Tier-II cities, and highway corridors continue to have limited charging coverage.

Unavailability of the Charging infrastructure where it is needed leads to discourage adaptation of EVs as compared to petrol vehicles.

This uneven distribution creates gaps in accessibility and discourages riders from undertaking longer trips.

2. Home Charging Limitations

A significant proportion of urban customers live in apartment complexes, rented accommodations, or gated communities where installing a private charger is difficult due to:

- Lack of dedicated parking.
- Shared electricity connections.
- Housing society restrictions.

- Absence of charging infrastructure.

As a result, many customers depend entirely on public charging infrastructure.

3. Peak-Hour Charger Congestion

Popular charging stations often experience high utilization during office commuting hours, weekends, and holidays.

Users currently have limited visibility regarding:

- Waiting time.
- Queue length.
- Charger occupancy.
- Estimated charger availability.

This increases uncertainty and negatively impacts customer experience.

How might Ather Energy leverage data, intelligent software, and strategic infrastructure expansion to create a seamless, reliable, and accessible charging ecosystem that minimizes range anxiety, improves charging convenience, optimizes infrastructure utilization, and enhances the overall customer experience while supporting the rapid growth of electric mobility across India?

3.3 Key User Pain Points

Charging Accessibility

Users prefer charging stations within a short distance of their daily routes rather than making significant detours.

Waiting Time

Customers perceive waiting as one of the biggest inconveniences, particularly when charger occupancy is unpredictable.

Apartment Charging Issues

Customers without dedicated parking often find home charging impractical despite owning an electric scooter.

Emergency Charging

Users value fast charging not for routine use but as a reliable safety net during unexpected situations.

3.4 Behavioral Insights

Analysis of EV user behavior reveals several important patterns:

Insight 1

More than 80–90% of charging sessions for personal EV users typically occur at home when home charging is available.

Public charging is primarily used for convenience or longer journeys rather than daily charging.

Insight 2

Users prefer integrated digital experiences.

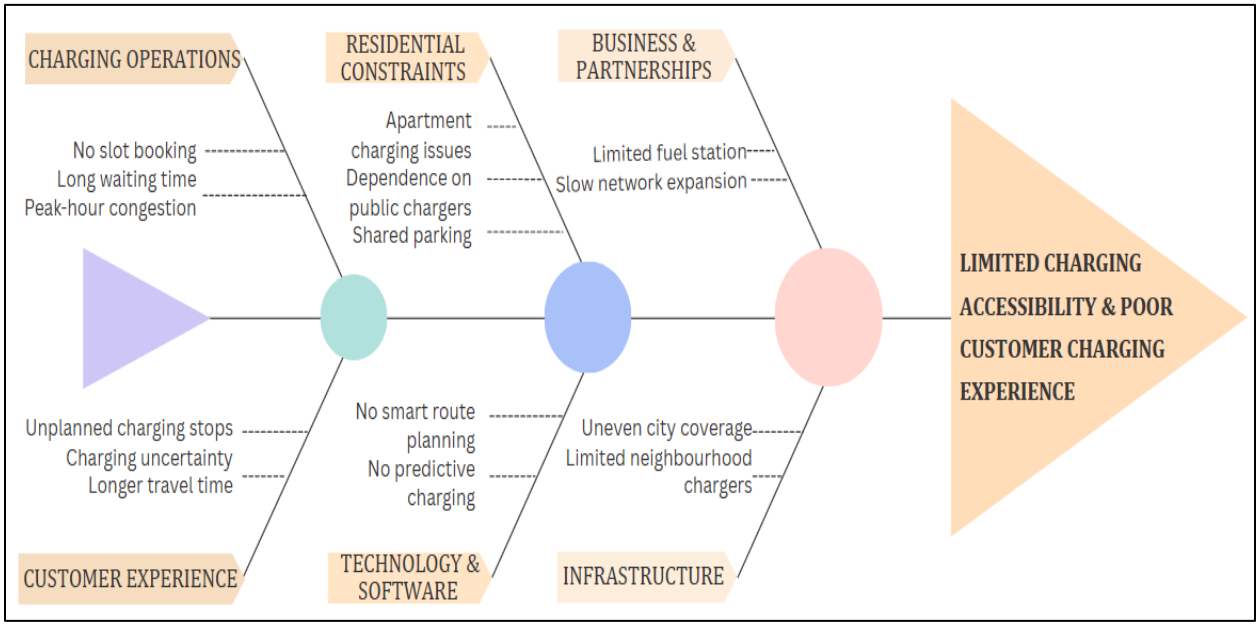
Rather than opening multiple applications for navigation, charging, and payments, customers expect a single ecosystem that handles trip planning, charger discovery, reservations, navigation, and payments seamlessly.

Insight 3

Customers increasingly value **time certainty** over charging speed.

Knowing that a charger will be available upon arrival often provides more reassurance than simply having access to a high-powered charger.

3.3.1 FISHBONE DIAGRAM



3.5 Opportunity Areas

Based on these insights, several high-impact opportunities emerge:

- AI-driven charging route planning with battery-aware navigation.
- Predictive charger recommendations based on rider behavior.
- Data-driven deployment of neighborhood Chargers in underserved areas.
- Premium charger reservation and queue management.
- Real-time occupancy prediction using historical demand.

- Subscription plans for heavy public charging users.
- Partnerships with residential societies, workplaces, and fuel retailers to accelerate infrastructure expansion.
- The new policy of Delhi government allows registration of EV 2-wheelers only from 1st April 2028 leads to opening gate of opportunities for Ather to expand and capture major portion of market of 2-whellers.

3.6 Product Management Insight

From a Product Management perspective, the core challenge is not simply increasing the number of charging stations but designing an ecosystem that reduces uncertainty at every stage of the charging journey. A successful charging experience depends on four interconnected pillars:

1. **Infrastructure Availability**- ensuring chargers are deployed where demand is highest.
2. **Digital Intelligence**- using real-time data and predictive algorithms to guide users to the right charger at the right time.
3. **Operational Efficiency**- minimizing wait times through reservations, occupancy forecasting, and dynamic utilization.
4. **Customer Trust**- creating confidence that charging will be accessible whenever and wherever it is needed.

By addressing these dimensions together, Ather can transform charging from a functional necessity into a strategic competitive advantage, strengthening customer loyalty, improving infrastructure utilization, and accelerating the adoption of electric mobility across India.

4. MARKET ANALYSIS & COMPETITOR BENCHMARKING

4.1 Indian Electric Two-Wheeler Market Overview

India is the world's largest two-wheeler market, with millions of scooters and motorcycles sold annually. Increasing fuel prices, stricter emission norms, government

incentives under schemes such as FAME II and state EV policies, and growing consumer awareness of sustainable mobility have accelerated the adoption of electric two-wheelers.

Over the last few years, electric scooters have evolved from being niche products to mainstream urban mobility solutions. Consumers now expect electric scooters to offer not only low running costs but also connected technology, dependable performance, and a convenient charging experience.

While battery technology has improved significantly, charging infrastructure remains one of the key differentiators among EV manufacturers. Customers increasingly evaluate brands based on:

- Availability of charging stations
- Charging speed
- Ease of locating chargers
- Route planning capabilities
- Charging costs
- Reliability of the charging network
- Overall ownership experience

As EV penetration increases beyond metropolitan cities into Tier II and Tier III markets, companies that build scalable and customer-centric charging ecosystems will gain a significant competitive advantage.

4.2 Government Policy Driving EV Adoption in Delhi

The Delhi EV Policy 2026 presents a significant growth opportunity for Ather Energy. Under the approved policy, only electric two-wheelers will be eligible for new registration in Delhi from 1 April 2028, while existing petrol two-wheelers can continue to operate. This policy is expected to accelerate the transition from internal combustion engine (ICE) scooters to electric vehicles, creating a substantial increase in demand for EV charging infrastructure.

Product Management Insight: Rather than treating the 2028 policy as merely a regulatory change, Ather should view it as a demand-generation event. Proactively expanding charging infrastructure, improving digital charging experiences, and securing strategic charging locations before the policy takes full effect can position Ather as the preferred charging ecosystem for the next wave of electric two-wheeler users in Delhi-NCR.

4.3 Competitor Comparison

Although Ather has developed one of the strongest charging ecosystems in India, several gaps remain between the existing customer experience and the ideal charging journey.

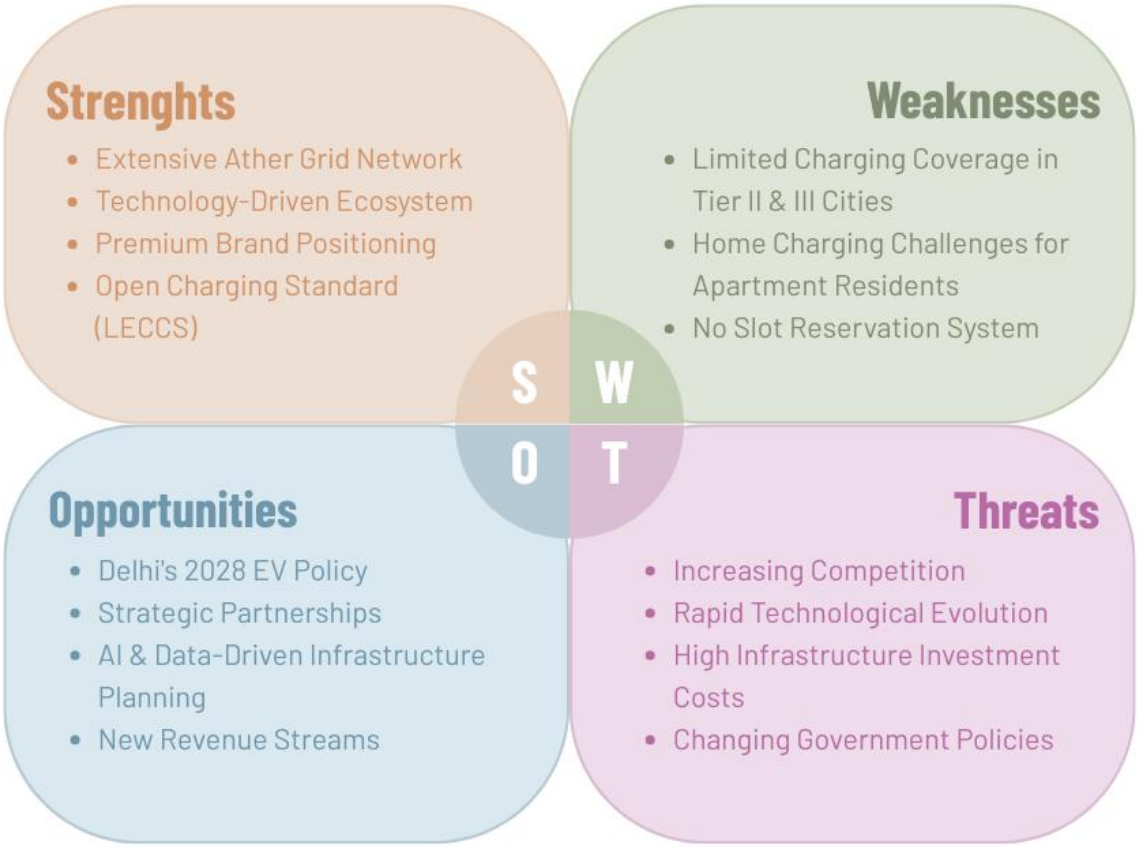
Parameter	Ather Energy	Ola Electric	TVS iQube	Bajaj Chetak	Hero Vida
Home Charging	Available	Available	Available	Available	Available
Fast Charging Network	Extensive Ather Grid	Hypercharge Network	Limited	Limited	Expanding
Mobile App Integration	Advanced	Advanced	Moderate	Moderate	Advanced
OTA Updates	Yes	Yes	Partial	Limited	Yes
Navigation	Built- In	Built- In	Available	Limited	Available
Charging Route Planning	Basic	Available	Limited	Limited	Basic
Slot Reservation	Not Available	Not Available	Not Available	Not Available	Not Available
Charging Subscription	Limited	Limited	No	No	No
Connected Ecosystem	Strong	Strong	Moderate	Moderate	Strong

Color	Significance
Green	Strong / Best-in-Class
Yellow	Moderate / Developing
Blue	Basic / Needs Enhancement
Red	Unavailable / Not Offered

4.4 Gap Analysis

Although Ather has developed one of the strongest charging ecosystems in India, several gaps still exist between the current customer experience and the ideal charging journey, particularly in areas such as real-time charger availability, seamless route planning, and advanced user-centric features that enhance convenience and reliability.

4.5 SWOT Analysis of Ather



5. PROPOSED SOLUTIONS

5.1. Establish a Dedicated Charging Infrastructure Analytics Team

5.1.1 Background

Ather possesses one of the richest mobility datasets in the Indian EV ecosystem. Every connected scooter continuously generates anonymous telematics data including:

- Vehicle location
- Home charging frequency
- Public charging frequency
- Battery State of Charge (SOC)

- Daily travel distance
- Frequently travelled routes
- Charging duration
- Idle time after charging
- Peak charging hours
- Areas where battery levels consistently become critically low

Instead of relying solely on manual surveys or customer requests, Ather can leverage these insights to make evidence-based infrastructure decisions.

5.1.2 Objective

Develop a dedicated cross-functional team responsible for identifying optimal locations for new charging stations through advanced data analytics, ensuring infrastructure expansion aligns with actual customer demand rather than assumptions.

5.1.3 Proposed Cross-functional Team

The team should consist of professionals from multiple domains.

Department	Responsibilities
Data Scientists	Analyze charging patterns and predict future demand
GIS & Mapping Specialists	Perform spatial analysis and identify high-demand zones
Product Managers	Convert analytical insights into actionable product features
Business Strategy Team	Evaluate commercial feasibility of new charging stations
Operations Team	Coordinate installation and maintenance
Software Engineers	Build internal dashboards and data pipelines
Customer Success Team	Collect qualitative feedback from users

5.1.4 Data Sources

The team should analyze multiple datasets simultaneously, Like:

Dataset	Analysis
Vehicle Data	• Daily distance travelled • Average battery consumption • Remaining battery before charging • Ride frequency • Vehicle density by location
Charging Behaviour	• Home charging frequency • Public charging frequency • Fast charging usage • Average charging duration • Peak charging timings
Geographic Data	• Residential density • Commercial areas • IT parks • Shopping malls • Apartment societies • Metro stations • Educational institutions
Customer Profile	• Apartment residents • Independent house owners • Fleet operators • Delivery executives • Office commuters

Analytical Models

The analytics team can utilize:

- Heat Map Analysis
- GIS Mapping
- Demand Forecasting
- Clustering Algorithms (K-Means)
- Predictive Analytics
- Route Density Analysis
- Charging Gap Analysis
- Machine Learning Demand Prediction

5.1.5 Expected Outcomes

The analysis will identify:

- Areas with highest unmet charging demand
- Locations where scooters frequently reach below 20% battery
- Residential clusters without home charging facilities
- Popular commuter corridors requiring fast chargers
- Apartment societies lacking charging infrastructure

These insights will allow Ather to prioritize infrastructure deployment with significantly higher utilization rates.

5.1.6 Customer Benefit

This initiative is particularly beneficial for:

- Apartment residents
- Gated communities
- Rental accommodations
- PG residents

- Customers without dedicated parking
- Users unable to install personal chargers

Rather than forcing these customers to depend solely on public charging stations located far away, Ather can strategically establish Neighbourhood Chargers near their residences.

5.2 SMART ROUTE PLANNING WITH INTELLIGENT CHARGING STOPS

5.2.1 Background

Long-distance travel remains one of the primary causes of range anxiety among electric two-wheeler users. Riders often struggle to estimate whether the available battery capacity is sufficient to complete both onward and return journeys.

Although charging stations are displayed within the application, customers currently need to manually identify suitable charging locations during their trip.

5.2.2 Objective

Introduce an AI-powered trip planning feature within the Ather App that proactively recommends charging stops based on the rider's destination, battery level, traffic conditions, and charger availability.

5.2.3 Proposed Feature ("Plan My Ride")

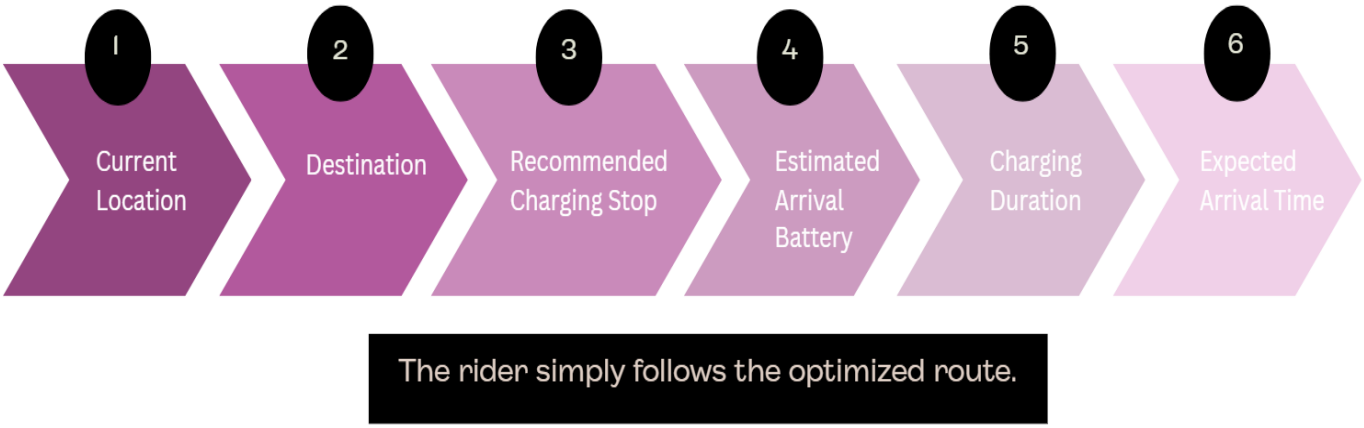
- The user enters:
- Destination
- Expected return trip
- Current battery percentage

The application automatically calculates:

Estimated battery consumption, Remaining range, Elevation, Traffic conditions, Weather impact, Charger availability .

The system then recommends:

- Charging stations
- Suggested charging duration
- Estimated arrival battery percentage
- Total trip time



5.2.4 Example Scenario

i. Journey

Ghaziabad → Gurgaon → Ghaziabad

Distance:

- One-way: 60 km
- Round Trip: 120 km

Current Battery:

65%

Instead of waiting until the battery becomes critically low, the application predicts:

"Battery will reach approximately 18% before returning home."

The application automatically suggests:

ii. Recommended Stop

Ather Grid

Location:

Sector 29 Gurgaon

Charging Duration:

18 minutes

Expected Battery After Charging:

70%

Additional Travel Time:

22 minutes

iii. Advanced Features

The features may also include:

- Real-time charger availability
- Live queue status
- Fastest charging route
- Lowest waiting time
- Multiple charging alternatives
- Traffic-adjusted battery prediction
- Weather-based range prediction

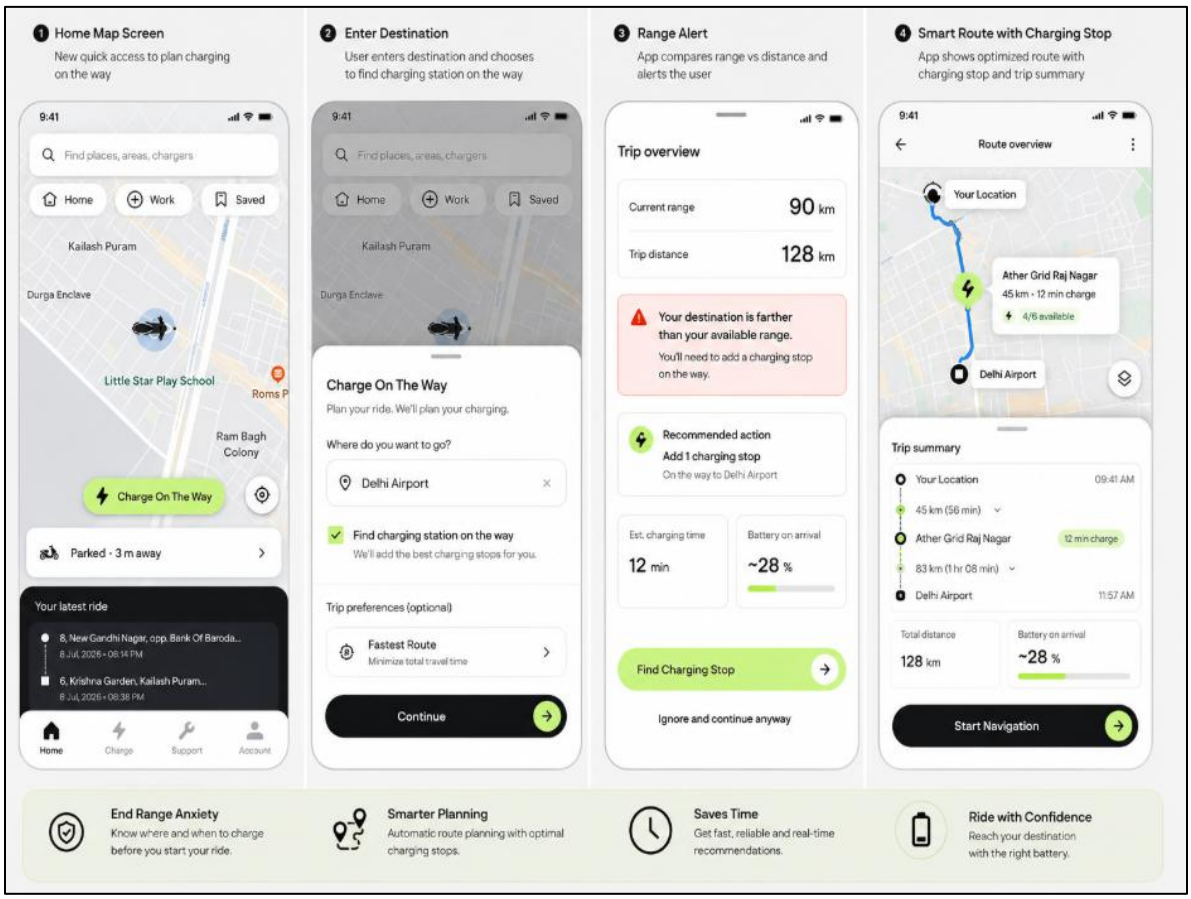
iv. Benefits

- Eliminates range anxiety
- Optimizes travel planning
- Improves charger utilization
- Increasing customer confidence
- Encourages longer rides

5.2.5 Prototype

Link- https://aistudio.google.com/apps/238fd53c-1aea-4de6-9d6d-1316c4127931?showPreview=true&showAssistant=true

High Fidelity diagram-



5. 3. Premium Charging Slot Reservation System

5.3.1 Background

During peak demand, customers often experience long waiting times due to limited charger availability. This particularly impacts commuters, emergency users, and riders undertaking long-distance trips.

5.3.2 Objective

Introduce a premium reservation system that enables customers to pre-book charging slots at eligible Ather Grid stations by paying a convenience fee.

5.3.3 Proposed Solution

Enable slot booking exclusively at charging stations equipped with three or more chargers, where one charger can dynamically function as a reservable charger.

Customers will be able to reserve charging sessions of:

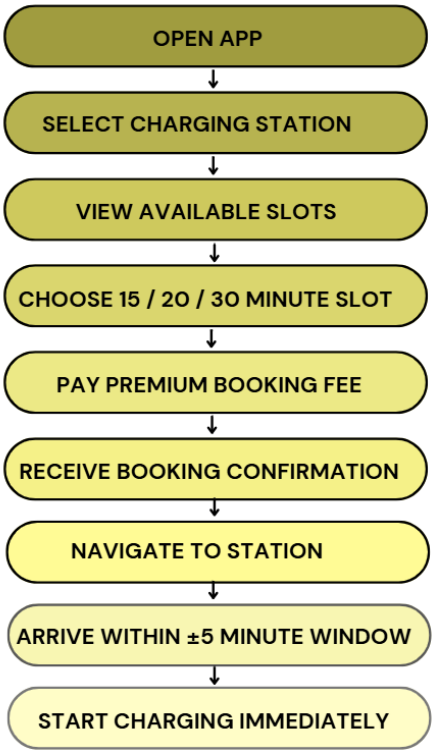
- 15 minutes
- 20 minutes
- 30 minutes

Users must arrive within five minutes before or after the scheduled booking time.

If the customer fails to arrive within the permitted time window, the reservation will automatically expire, and the charger will become available for public use.

To maximize charger utilization, reserved chargers will automatically function as normal public chargers whenever no active reservations exist.

5.3.4 Reservation Flow



5.3.5 Expected Benefits

- Reduced waiting time
- Guaranteed charger availability
- New premium revenue stream
- Improved customer experience
- Higher charger utilization

Success Metrics (KPIs)

Customer KPIs

KPI	Target
Average waiting time	↓ 40%
Booking success rate	>90%
Customer Satisfaction (CSAT)	>4.5/5
Charging experience rating	↑ 25%

KPI	Target
Repeat booking users	>35%
Queue abandonment rate	↓ 50%

5.3.6 Business Benefits (Ather)

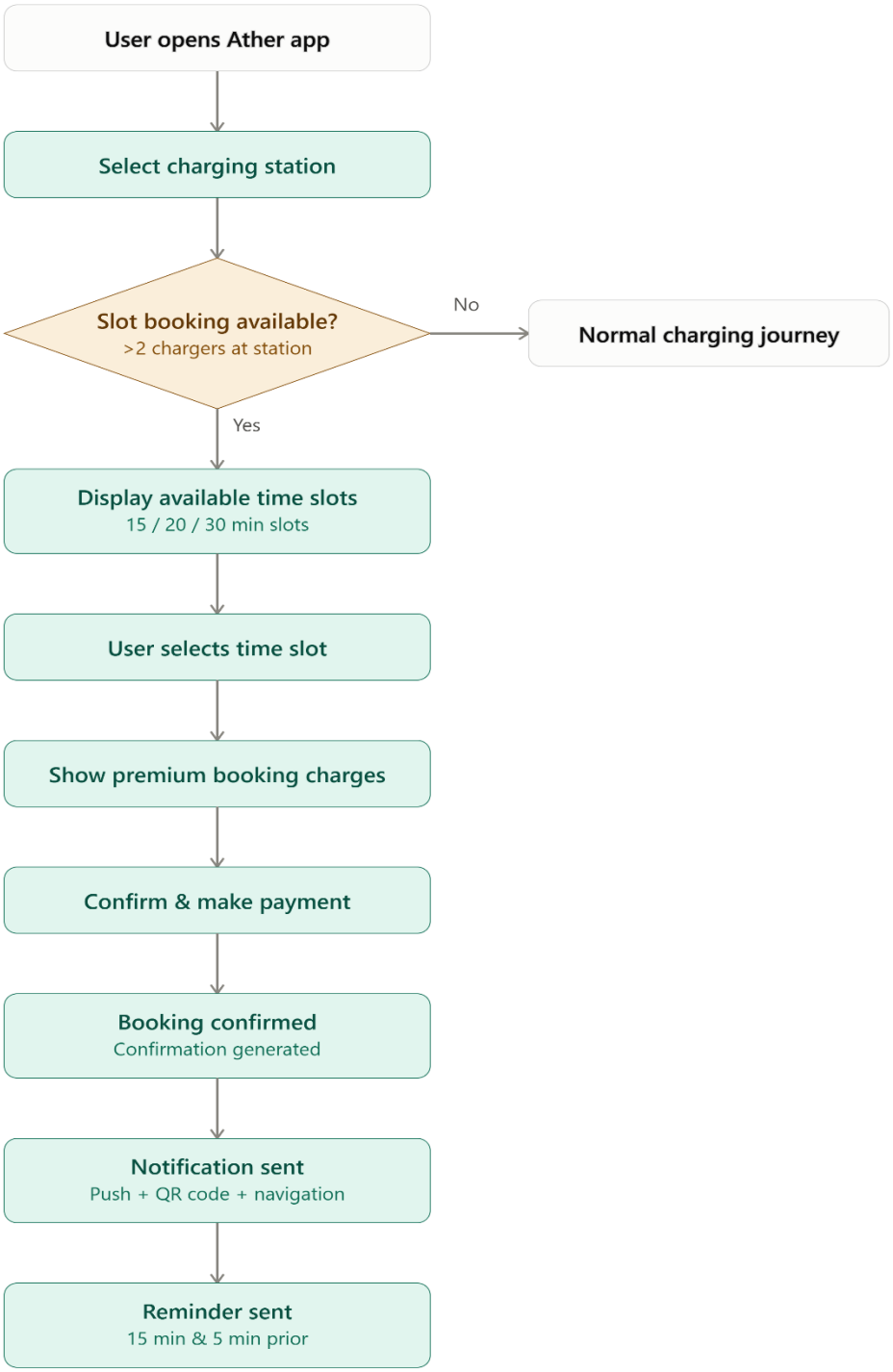
Benefit	Business Impact
New premium revenue stream	Booking fee generates additional income
Better charger utilization	Dynamic allocation minimizes idle chargers
Higher customer retention	Improved charging experience increases loyalty
Increased app engagement	More users interact through the app
Competitive differentiation	Unique feature among charging networks
Better demand forecasting	Slot data improves capacity planning
Reduced crowd management	Organized charging reduces station congestion

Business KPIs

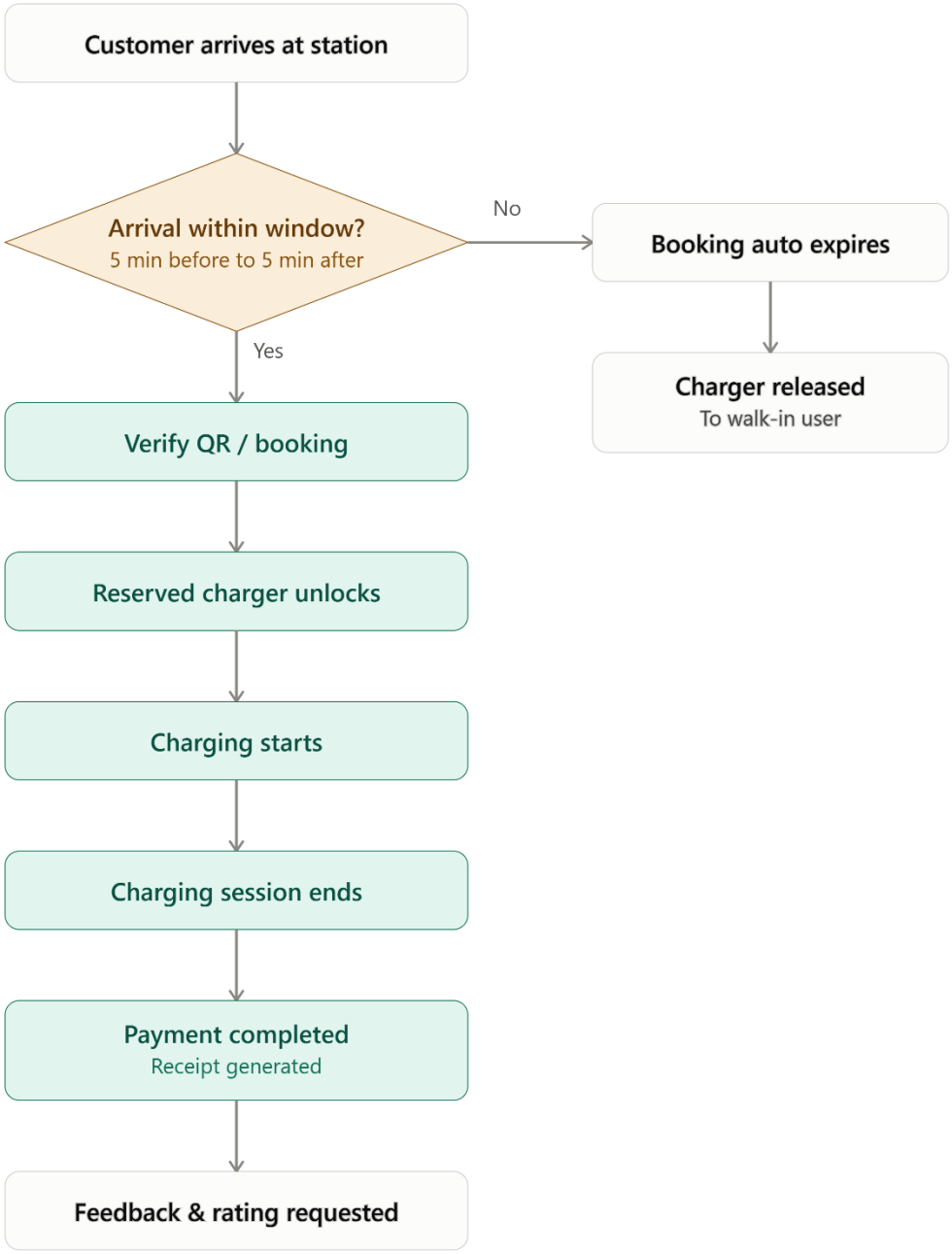
KPI	Target
Premium booking adoption	20–30% of eligible charging sessions
Incremental revenue from booking fees	Positive monthly growth
Charger utilization	>85%
App engagement	+20% charging-related sessions
Monthly Active Charging Users	Increase by 15%
Customer retention	Increase by 10%
Revenue per charging station	Increase by 8–12%

5.3.7 Premium Slot Booking Feature- Workflow

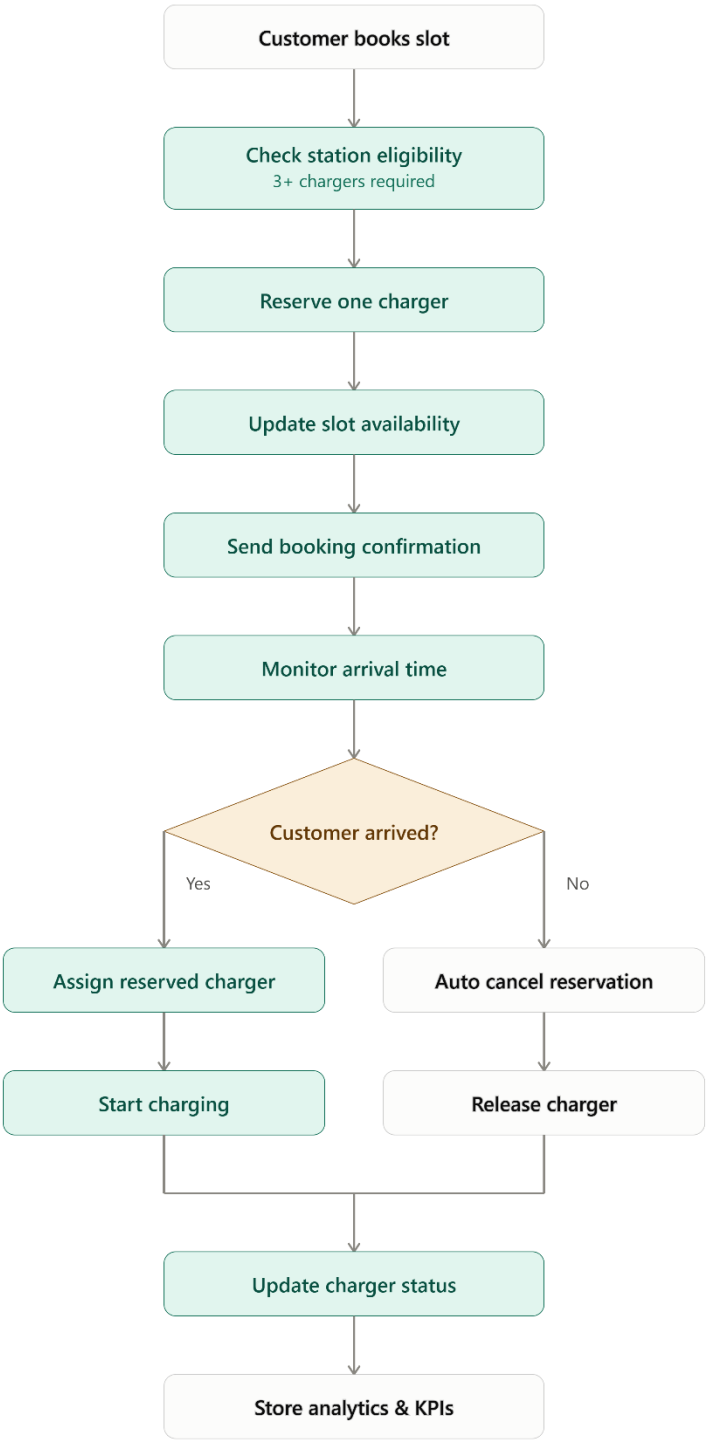
Given how many steps are here, let's split this into two diagrams the **booking flow** (open app through reminder) and the **arrival & charging flow** (arrival through feedback):



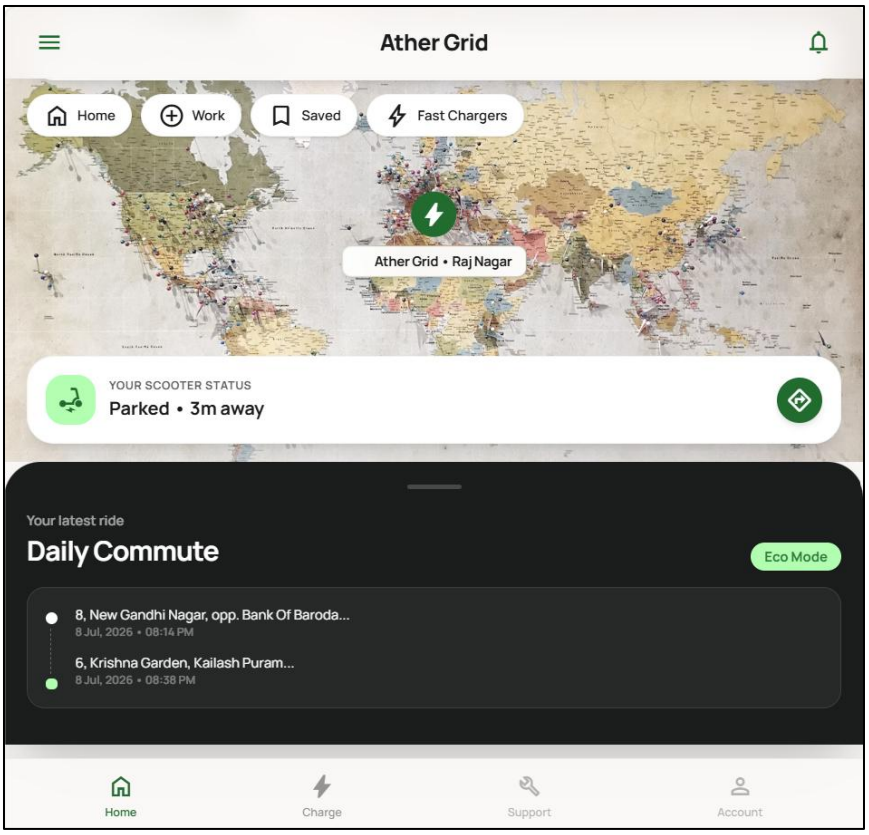
Here, the customer arrives at the station and the flow branches again based on whether they show up inside the allowed arrival window.

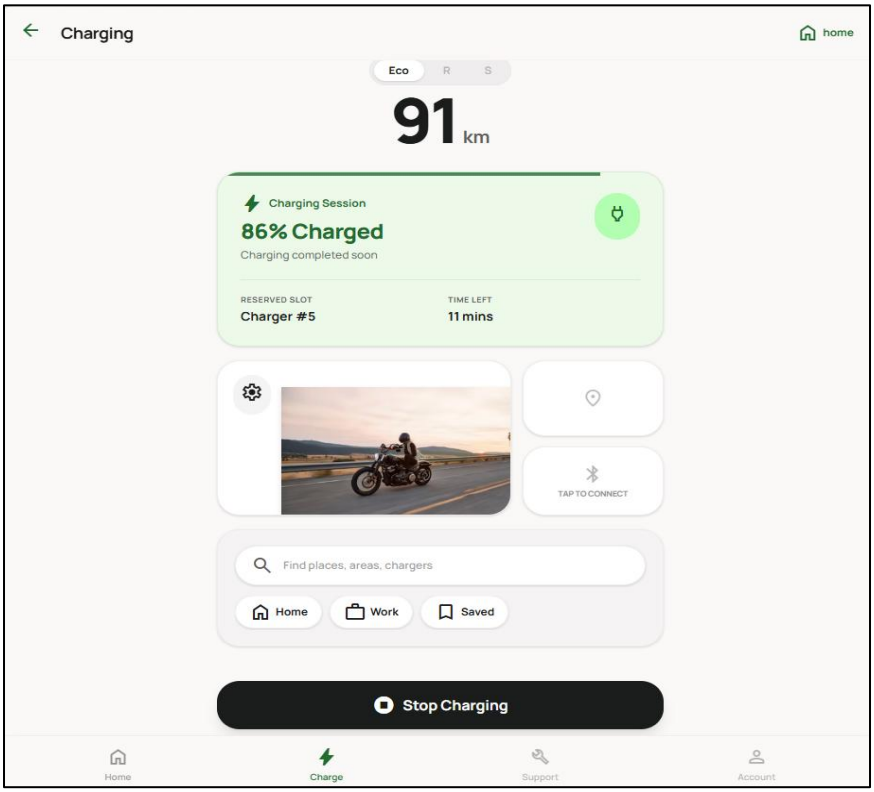


Backend Workflow



Prototype link - https://ather-grid-677520987911.asia-southeast1.run.app





6. MORE SOLUTIONS (MISCELLANEOUS)

6.1 STRATEGIC PARTNERSHIPS FOR RAPID CHARGING NETWORK EXPANSION

6.1.1 Objective

Accelerate Ather Grid expansion through strategic partnerships with fuel retailers, commercial establishments, and charging infrastructure providers.

6.1.2 Industry Context

Charging infrastructure remains one of the most significant barriers to widespread EV adoption. While Ather has established one of India's largest two-wheelers charging networks, expanding independently requires substantial capital investment and longer deployment timelines.

Ather has already strengthened its ecosystem by introducing the Light Electric Combined Charging System (LECCS), which has been recognized as India's standard charging connector for light electric vehicles. This open standard enables interoperability with multiple manufacturers and charging network operators.

6.1.3Proposed Solution

Expand infrastructure through partnerships with:

- Indian Oil Corporation (IOC)
- Bharat Petroleum Corporation Limited (BPCL)
- Hindustan Petroleum Corporation Limited (HPCL)
- Shopping malls
- Office parks
- Hotels
- Educational institutions
- Apartment developers

- Fleet operators

Installing Ather Grid chargers at these high-footfall locations will significantly improve accessibility while reducing infrastructure deployment costs.

The strategy can also leverage existing interoperability partnerships with Hero Vida, Matter, Bolt, Kazam, and EV amp to further strengthen the charging ecosystem.

6.1.4 Expected Benefits

- Faster charging network expansion
- Lower capital expenditure
- Increased charger visibility
- Higher charging accessibility across Tier I, Tier II, and Tier III cities
- Stronger ecosystem partnerships

6.2ATHER CHARGING+ SUBSCRIPTION

6.2.1Objective

Introduce a subscription-based charging membership designed for customers who frequently rely on public charging infrastructure.

6.2.2Background

Daily commuters, gig workers, and customers without access to home charging incur recurring expenses, making predictable pricing an attractive value proposition.

6.2.3Proposed Solution

Launch **Ather Charging+**, a subscription model offering discounted or unlimited charging benefits over a fixed subscription period.

A subscription model can reduce their operational costs while increasing customer loyalty.

Plan	Price	Benefits
Basic	Monthly	Discounted charging rates
Standard	Monthly	Higher charging limits, priority support
Premium	Monthly	Unlimited charging (Fair Usage Policy), priority access, reservation discounts

Target Audience

- Delivery executives
- Fleet operators
- Daily commuters
- Ride-sharing users
- Customers without home charging

6.2.4 Expected Benefits

- Increased customer retention
- Predictable recurring revenue
- Higher public charger utilization
- Enhanced value for premium customers
- Greater adoption among frequent riders

6.3. Data-Driven Marketing and Investment Outreach

6.3.1 Objective

Leverage location intelligence and targeted digital marketing to accelerate charging infrastructure expansion through local investment partnerships.

6.3.2 Background

Certain regions exhibit growing EV adoption despite limited charging infrastructure. Identifying these underserved markets presents an opportunity to expand the charging network through collaborative investment models.

6.3.3 Proposed Solution

Utilize insights generated by the analytics team to identify high-demand regions lacking adequate charging infrastructure.

Launch location-specific digital marketing campaigns targeting:

- Commercial property owners
- Fuel station operators
- Shopping centres
- Apartment associations
- Local entrepreneurs
- Potential infrastructure investors

Campaigns should communicate the commercial potential of hosting Ather Grid charging stations while highlighting the increasing demand for public EV charging.

6.3.4 Expected Benefits

- Faster deployment of charging stations
- Increased private-sector participation
- Lower infrastructure investment burden
- Improved charging accessibility in underserved regions
- Stronger regional brand presence

7. PRODUCT STRATEGY SUMMARY

Problem

EV adoption is growing rapidly, but **charging accessibility, range anxiety, and unpredictable waiting times** remain key barriers. While Ather has built a strong charging network, customers still face challenges in planning long journeys and accessing chargers during peak demand.

Solution

This project proposes a comprehensive charging ecosystem consisting of five initiatives:

- **Charging Intelligence Team** for data-driven charger expansion.
- **Smart "Charge on the Way"** feature for AI-based route planning.
- **Premium Slot Booking** to guarantee charger availability during urgent situations.
- **Ather Charging+ Subscription** for frequent users.
- **Strategic partnerships** with fuel stations and commercial properties to accelerate charging network growth.

Together, these solutions enhance customer experience while improving business performance.

8. PRODUCT MANAGEMENT PERSPECTIVE

This project is not about building isolated features- it is about solving the complete charging journey.

PM Principle	How This Project Addresses It
User-Centricity	Reduces range anxiety and waiting time.
Data-Driven Decisions	Uses charging and location data to plan infrastructure.
Business Growth	Introduces premium booking and subscriptions as new revenue streams.
Operational Efficiency	Improves charger utilization with dynamic allocation.
Scalability	Leverages partnerships and AI to support future growth.

9. SUCCESS METRICS

The effectiveness of the product will be measured through:

Customer Metrics

- 40% reduction in average waiting time.
- 25% improvement in customer satisfaction (CSAT/NPS).
- 30% increase in successful long-distance trips.
- Higher adoption of "Charge on the Way" and Slot Booking features.

Business Metrics

- 20–30% adoption of premium slot booking.
- 10–15% increase in charging-related revenue.
- More than 85% charger utilization.

- 15% increase in monthly active users of the Ather app.
- 10% improvement in customer retention.

10. RISKS AND MITIGATION

Risk	Mitigation
Low adoption	Introductory offers and user education.
High no-show rate	Auto-cancel after the grace period and optional penalties.
Under-utilized reserved chargers	Dynamic release during off-peak hours.
Infrastructure expansion costs	Strategic partnerships with fuel stations and commercial properties.

11. PRODUCT ROADMAP

Short Term (0–6 Months):

- Build Charging Intelligence Team.
- Launch Smart Route Planner.
- Pilot Premium Slot Booking.

Medium Term (6–12 Months):

- Launch Ather Charging+ Subscription.
- Expand Neighborhood Chargers.
- Introduce AI-based occupancy prediction.

Long Term (1–3 Years):

- Nationwide partnerships with IOC, BPCL, and HPCL.
- AI-driven demand forecasting.
- Universal interoperable charging ecosystem across India.

12. FINAL RECOMMENDATION

Ather should move beyond being an EV manufacturer and position itself as **India's most intelligent charging ecosystem**. By combining digital innovation, data-driven infrastructure planning, premium services, and strategic partnerships, the company can reduce charging uncertainty, improve customer trust, and create sustainable business growth.

13. CONCLUSION

Charging infrastructure is no longer just a support service it is a core part of the electric mobility experience. As EV adoption accelerates, customers expect charging to be as reliable and predictable as refueling a conventional vehicle.

This project demonstrates how Ather can transform the charging experience through intelligent software, data-driven infrastructure expansion, premium slot booking, and strategic partnerships. These initiatives not only address current customer pain points such as range anxiety and waiting time but also create new revenue opportunities, improve operational efficiency, and strengthen customer loyalty.

From a Product Management perspective, the success of these solutions lies in delivering measurable customer value while achieving sustainable business outcomes. By continuously monitoring key metrics, iterating based on user feedback, and scaling successful initiatives, Ather can build a future-ready charging ecosystem that accelerates EV adoption and reinforces its leadership in India's electric mobility market.

REFERENCES

1. Ather Energy. (2025). Annual report 2024–25. https://www.atherenergy.com
2. Ather Energy. (2024). Impact report 2024. https://www.atherenergy.com
3. Ather Energy. (n.d.). Ather Energy official website. https://www.atherenergy.com
4. Bansal, P., Kumar, R. R., Raj, A., Dubey, S., & Graham, D. (2021). Willingness to pay and attitudinal preferences of Indian consumers for electric vehicles. arXiv. https://arxiv.org/
5. Bureau of Indian Standards. (n.d.). Light Electric Combined Charging System (LECCS) standards. https://www.bis.gov.in
6. Chen, Y., Wu, G., Sun, R., Dubey, A., Laszka, A., & Pugliese, P. (2020). A review and outlook of energy consumption estimation models for electric vehicles. arXiv. https://arxiv.org/
7. Figma. (n.d.). Figma. https://www.figma.com
8. Google. (n.d.). Google Maps Platform documentation. https://developers.google.com/maps/documentation
9. International Energy Agency. (2026). Global EV outlook 2026. https://www.iea.org/reports/global-ev-outlook-2026
10. International Energy Agency. (n.d.). Global EV data explorer. https://www.iea.org/data-and-statistics/data-tools/global-ev-data-explorer
11. Ministry of Heavy Industries, Government of India. (n.d.). FAME India Scheme and PM E-DRIVE Scheme. https://heavyindustries.gov.in
12. Ministry of Road Transport and Highways. (n.d.). VAHAN dashboard. https://vahan.parivahan.gov.in
13. NITI Aayog. (n.d.). National electric mobility mission and electric vehicle initiatives. https://www.niti.gov.in
14. OpenAI. (2025). ChatGPT. https://chat.openai.com
15. Society of Indian Automobile Manufacturers. (n.d.). Industry statistics and reports. https://www.siam.in
16. Government of National Capital Territory of Delhi. (n.d.). Delhi electric vehicle policy. https://ev.delhi.gov.in

CONTRIBUTION –

Bhavika Bharadwaj

- Developed user stories, acceptance criteria, assumptions, constraints, and implementation considerations.
- Ensured the solutions were aligned with user needs and business goals.

Sneha Tiwari

- Designed all flowcharts, process diagrams, workflow illustrations, KPIs.
- Created the Fishbone Diagram for root cause analysis.
- Structured the problem statements, objectives, proposed solutions, and implementation framework.
- Prepared the complete project documentation.
- Edited and standardized the document formatting, content flow, tables, and graphical elements.

Vardaan Mahendra

- Team Formation and partnership.
- Identified and formulated the project's problem statements through secondary research and customer pain-point analysis.
- Conducted ideation and conceptualized innovative solutions to improve Ather's charging ecosystem.
- Designed and developed interactive prototypes for the proposed product features.
- Created high-fidelity wireframes and prototype screens to demonstrate feature functionality and user experience.